

Neel Shah

ML & Computer-Vision Engineer · Quantitative Modeling

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SUMMARY

ML / computer-vision engineer with deep quantitative-modeling chops. I built a **multi-sport prediction platform** end to end — **five sports priced by one calibrated engine**: broadcast-video CV tracking → a possession-level Monte-Carlo simulator pricing **~372 markets per game** → a **live in-play repricer with proven calibration and a forward-CLV ledger** — plus the **walk-forward / leak-detection harnesses that validate (and disprove) my own results**. It runs itself (**18+ self-healing services**, unattended daily paper loop) and ships a **public MCP server** so anyone can query it from their own Claude. The measured win: **in-game conditioning** — fusing the pregame rating prior with realized mid-game state cuts win-probability Brier from **0.209 to 0.159 (NBA)** and **0.241 to 0.126 (MLB)**, leak-free and out-of-sample. CV tracking runs at **~\$0.10/game**. Against real closing lines the market is **efficient**, and every candidate pregame edge was retracted by my own gates. My own harness caught my early "+18%" as an artifact and I retired it. B.S. Data Science in progress.

SELECTED PROJECT — COURTVISION

Multi-sport prediction platform: broadcast-video CV → calibrated Monte-Carlo pricing engine → live in-play repricing → autonomous paper execution, with the validation harness as a first-class component. github.com/neelshah/court-vision · evidence: [docs/JOB_EVIDENCE_PACKET.md](#) · [docs/IN_GAME_SHOWCASE.md](#)

Platform — one engine, five sports

- **Sport-blind kernel**: NBA, MLB, club soccer, World Cup, and tennis all speak the same **predict / predict_live** interface; one possession-level **Monte-Carlo sim coherently prices ~372 markets per game** (ML/totals/spreads/1X2, player props, alt ladders, SGPs) off a single calibrated anchor.
- **Live + autonomous**: in-play calibration **proven on held-out live snapshots**; every paper bet graded by **forward CLV** in a transactional ledger with off-market-price guards; **18+ self-healing supervised services** run the daily loop unattended; **public MCP server + one-command data pack** lets reviewers interrogate the system from their own Claude.

Edge — validated, and precisely scoped (the differentiator)

- **In-game conditioning (the measured, leak-free win)**: fusing the pregame rating prior with realized mid-game state sharpens the win-probability forecaster — **Brier 0.209 → 0.159 on NBA, 0.241 → 0.126 on MLB**, real-corpus out-of-sample. This is a *calibration* win, not a \$ edge: a live book sees the same score. A separate in-play assists projection beats the live line ~66% of the time, but that corpus is ~2 games, so I report it as accuracy only.
- **Pregame (the honest negative result)**: vs real DK/FanDuel/MGM closing lines the model is **break-even-minus-vig** (~-2%). The market is efficient, and every candidate edge, including my strongest (assists), was **retracted by my own gates**.
- **Rigor (the differentiator)**: my walk-forward / leak-detection harness **caught my early "+18% ROI" headline as a market-follow artifact** — and a Q4 data-leak in an in-play metric — and I **retired both and published the negative results**. I price an edge, then build the instruments to disprove it.

Computer Vision

- Full **broadcast-video → court-coordinate** pipeline (YOLOv8 detection → SIFT homography → Kalman+Hungarian tracking → OSNet re-ID) on a single consumer GPU, **~\$0.10/game** vs six-figure Sportradar / Second Spectrum licensing.
- Tracker built from primitives — **6-D Kalman + Hungarian** over blended IoU+appearance cost; custom **YOLOv8n** ball detector (ONNX/TensorRT); **OSNet** re-ID reimplemented in PyTorch; replay/scene-cut-hardened **homography**; slots resolved to real NBA identities across **240+ games**.

ML Engineering & Validation

- **Walk-forward (expanding-window)** CV with a per-fold **leakage guard** + CI gate that fails the build on overfitting; **truncation-invariance** property tests for lookahead bias.
- Per-stat **XGBoost** with **isotonic** post-calibration; four de-vig methods from scratch incl. **Shin (1992)**; **fractional-Kelly** with correlation penalty + drawdown breaker; multi-corpus acceptance gate (ship only if it beats baseline on ≥2 independent OOS corpora).

Backend & Systems · Data Engineering

- **FastAPI** serving layer (**~99 endpoints**, REST + WebSocket + SSE) with watchdog-supervised daemons; PostgreSQL→SQLite fallback persistence; transactional P&L ledger; **CI/CD**; 5 Docker images; **~7,400-test** pytest suite.
- **Agentic research loop**: Claude agents autonomously discover, validate, ship, and retire signals via an ~80-artifact intelligence layer between CV tracking and the models.
- **291K-row** player-vs-player matchup DB from **2,214** raw tracking files across 3 seasons; idempotent single-writer merge of ~80 derived artifacts into a **690-node** knowledge graph with point-in-time-correct (shift-1) joins.

PROFESSIONAL EXPERIENCE

Data Analyst Intern — Sunsolor · Tempe, AZ · 2025

Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist — DataAnnotation.Tech · Remote · 2024–2025

Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored edge-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst — Fortrex Securities · Madison, WI · 2023–2024

Built secure executive reporting interfaces (99.9% uptime); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

EDUCATION & CERTIFICATIONS

B.S. Data Science — University of Iowa · 2022–2027, **expected May 2027** (3 years coursework completed; finishing remaining credits online). Coursework: math stats, probability theory, machine learning, financial analytics.

Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

Languages: Python (expert), SQL, JavaScript/TypeScript (working)

ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels

Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau

Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration